

Group Name: 0110 0111

Group Members: Chang Yi Hern, Bryan Tan Chun Xing, Chin Yoong Khang, Wong Ming Ze

Chosen IP:

Method for Estimating Chronological Age from Dental Orthopantomogram Images Using a Convolutional Neural Network

Technology Overview — How the AI Works

What is an OPG?

A dental orthopantomogram (OPG) is a panoramic X-ray image that captures the full mouth in a single scan — all teeth, both jaws, surrounding bone structures, and developing tooth buds. It is a standard, non-invasive imaging tool already used routinely in dental clinics, hospitals, and forensic laboratories worldwide. No additional radiation exposure or specialised equipment is required beyond what already exists in any dental facility.

What biological signals does the AI read?

An OPG is not just a picture of teeth — it is a biological timeline. The structures visible in an OPG change in predictable, measurable ways as a person ages, across two distinct phases:

In younger individuals (approximately ages 5–21), the primary age signal comes from **tooth development stages**. Each tooth follows a known developmental sequence — crown formation, root elongation, apex closure — that can be precisely correlated to chronological age. The order and timing of tooth eruption, particularly the second molars and wisdom teeth, narrows the age estimate significantly. A trained expert — or a trained AI — reads these stages the same way a growth chart reads height.

In adults (ages 21 and above), the developmental signal fades and the AI shifts to reading **degenerative markers**: secondary dentin deposition (visible as a narrowing of the pulp chamber over time), root dentine translucency, alveolar bone height changes, and enamel wear accumulation. These markers degrade at statistically consistent rates across populations, making them reliable predictors of age even in older individuals where tooth development is complete.

How does the ResNeXt-50 CNN produce an age output?

Traditional manual methods require a forensic specialist to visually score each of these markers one by one using a reference atlas (such as Demirjian's method for children or Kvaal's method for adults), then calculate an estimated age — a process that takes 2–4 hours per case and varies significantly between practitioners.

Our system replaces this manual scoring process entirely using a ResNeXt-50 convolutional neural network. ResNeXt-50 was originally pre-trained on ImageNet, a dataset of over 14 million general images, which means it already knows how to detect edges, textures, gradients, and structural patterns in images before it ever sees a single dental X-ray. This is called **transfer learning** — rather than training from scratch, we take this pre-trained foundation and fine-tune its final layers specifically on Malaysian OPG images where the true chronological age is already known.

The model does not classify age into broad categories (e.g. "10–20 years"). Instead it performs **age regression** — it outputs a single continuous number representing the predicted chronological age. This is clinically and legally more useful because it gives forensic officers and immigration authorities a specific, defensible estimate rather than a wide bracket.

How is accuracy achieved?

Accuracy in our system comes from three compounding factors:

The first is **population specificity**. Most existing AI age estimation tools were trained on European or mixed-population datasets. Dental development timing, eruption patterns, and wear rates vary by ethnicity and diet. A model trained on Turkish or Belgian OPGs will produce higher error rates when applied to Malaysian patients. Our model is fine-tuned specifically on a Malaysian multi-ethnic dataset covering Malay, Chinese, and Indian demographics — the three major population groups encountered in PDRM and JIM caseloads. This is the single most important reason our accuracy target of MAE below 2 years is achievable where generic global tools fall short on Malaysian cases.

The second is **standardised preprocessing**. Before any OPG image enters the model, it passes through a preprocessing pipeline that normalises image dimensions, pixel intensity, and colour channels to a consistent format. This removes the variability introduced by different OPG machines across different facilities — whether the image comes from a hospital in KL or a detention centre in Sabah, the model receives a consistent input.

The third is **data augmentation during training**. The training pipeline artificially expands dataset variability by applying controlled transformations — rotation, flipping, brightness adjustment — to existing OPG images. This teaches the model to generalise across real-world variation in image quality and patient positioning, rather than memorising a narrow training set.

What does a Mean Absolute Error (MAE) of under 2 years mean in practice?

MAE is the average difference between the AI's predicted age and the person's true age across all test cases. An MAE of under 2 years means that if the system predicts a person is 25 years old, the true age is on average within ± 2 years — placing them between 23 and 27. For the forensic and immigration context where the critical legal threshold is 18 years of age (determining juvenile vs. adult classification), an MAE of under 2 years provides a reliable, defensible margin. A person predicted at age 14 is clearly juvenile; a person predicted at age 23 is clearly adult. The ambiguous zone is narrow and documented, which is more transparent and consistent than the current manual process where two specialists examining the same OPG can produce estimates that differ by 3–5 years with no audit trail.

End-to-end within 3 minutes

The full process — from OPG image upload to a court-ready PDF report — is designed to complete within 3 minutes without any AI expertise required from the user. A forensic officer uploads the image through the platform interface, the preprocessing pipeline standardises it, the ResNeXt-50 model runs inference, and the system generates a structured report containing the predicted age, confidence range, and the key dental markers identified — formatted for direct use in legal proceedings.

Value Proposition Statement:

We help PDRM forensic laboratories and immigration officers who struggle with days-long, subjective age estimations for undocumented individuals by offering an AI-powered dental OPG prediction system that delivers automated, highly accurate chronological age results in 3 minutes, enabling faster legal classification and reducing processing delays.

Problem Statement:

In Malaysia, forensic and legal cases involving undocumented individuals require accurate age estimation, but current methods rely on specialist radiologists, take 5–10 days per case, and produce inconsistent results. This

delays case processing and creates legal risk from age misclassification. Traditional methods are slow (expert-dependent manual assessment of dental images), highly subjective (results vary by practitioner), and unreliable (no standardised benchmarks). PDRM forensic units and Jabatan Imigresen Malaysia (JIM) both face high-volume caseloads where processing speed and accuracy directly impact legal proceedings, deportation decisions, and case closure timelines. Malaysia lacks an automated, accurate, and scalable AI-based solution for chronological age estimation that meets both forensic and immigration standards.

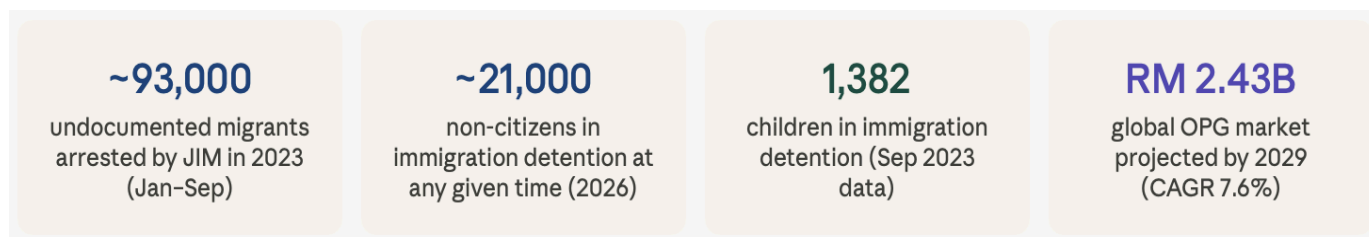
One-Line Pitch:

Manual age estimation takes days and stalls immigration cases but our AI reads dental x-rays in 3 minutes, enabling faster legal classification of undocumented individuals.

Target User:

Primary Target: Forensic laboratories (PDRM)

Malaysia handles thousands of undocumented migrant and juvenile cases yearly. Age is legally critical — current method is slow, costly, subjective. The budget exists within MOI/PDRM. The government procurement pathway is real. With over 93,000 arrests between Jan–Sep 2023 and approximately 21,000 non-citizens held in detention as of 2026, the volume of cases requiring age verification is immense — each one represents a potential juvenile at risk of legal misalignment.



Market Opportunity

- TAM ~RM 500M (ASEAN forensic + immigration age verification)
- SAM ~RM 50M ARR (Malaysia institutional pipeline, Years 1–4)
- SOM RM 8.5M ARR by Year 5.

TAM — Total Addressable Market

Forensic dental age estimation in ASEAN governments. Malaysia alone processes 80,000–100,000+ undocumented migrant cases annually via JIM enforcement operations, with an estimated 15–30% requiring formal age verification for juvenile determination. Across ASEAN (Indonesia, Thailand, Philippines, Singapore), the forensic immigration caseload exceeds 500,000 annually. Global OPG AI diagnostics market projected at RM 11B+ by 2030 (CAGR 7.6%). TAM: ~RM 500M (government forensic + immigration age verification, ASEAN).

SAM — Serviceable Addressable Market

Malaysia forensic + immigration + dental chain market reachable within 3 years. PDRM + JIM + MOH district hospitals (primary pipeline) + private dental chains. Estimated 80+ government institutional sites and 200+ private dental clinic targets in Malaysia. SAM: ~RM 50M ARR (Malaysia, Years 1–4 horizon).

SOM — Serviceable Obtainable Market

Realistically capturable by Year 5 with MDA registration and ASEAN expansion. 120+ direct sites + 2 white-label partners + API marketplace. Matches the RM 8.5M ARR target in the roadmap. SOM: RM 8.5M ARR by Year 5 — directly tied to the existing revenue model.

Secondary Target: Legal Authorities (Jabatan Imigresen Malaysia — JIM)

JIM is Malaysia's frontline agency for immigration enforcement and detention management. Operationally, JIM detention centres process thousands of undocumented individuals annually, many of whom carry no official identification or birth documentation. Age determination in these cases is not administrative — it is legally consequential. An individual classified as a juvenile is subject to entirely different legal proceedings, detention conditions, and deportation protocols than an adult. Getting this wrong exposes JIM to legal challenge, international scrutiny, and wrongful detention liability.

Currently, JIM relies on referrals to external forensic specialists for age estimation, adding 5–10 days to case processing per individual. With approximately 21,000 non-citizens held in detention as of 2026 and enforcement operations generating 80,000–100,000+ cases annually, this bottleneck compounds across the entire detention pipeline. JIM has both the operational urgency and the institutional budget — through MOI procurement channels — to adopt a faster, auditable, and court-ready solution. Our system reduces per-case turnaround from days to 3 minutes and produces a standardised PDF report with documented methodology, directly addressing JIM's need for defensible, legally admissible age evidence.

Tertiary Target: Private Dental Clinics

Private dental clinics represent a different use case from the forensic and immigration channels — the pain here is clinical and commercial rather than legal, but the output is the same: a court-ready, professionally formatted age estimation report.

For **forensic support**, private dental clinics are increasingly called upon to provide documentation for legal proceedings involving age disputes — particularly in family law, child protection, and immigration cases referred through private legal channels. Today, most private practitioners cannot fulfill these requests without referring the case to a forensic odontology specialist, losing both the billable work and the client relationship. Our system enables any clinic practitioner to generate an MDA-compliant, standardised age estimation report without forensic specialisation — opening a new billable service line that previously did not exist for most private dental practices. At RM 800–4,000 per manual specialist case versus RM 25 per case on our Professional tier subscription, the cost advantage is immediately legible to a clinic owner evaluating the ROI.

For **orthodontic treatment planning**, accurate age estimation from OPG images helps dentists determine whether a patient's skeletal and dental development is appropriate for their reported age — informing decisions on braces timing, jaw surgery candidacy, and growth modification treatments. This is a secondary but recurring use case that drives consistent monthly OPG reads, keeping utilisation high and churn low for Starter tier subscribers.

The private dental segment operates on shorter procurement cycles than government — no tender process, no ethics board approval required for software tools — making it the fastest channel to activate once MDA registration is complete. Target entry through dental distributor partnerships at the Starter tier (RM 999.90/month) from Year 3 onwards.

Sector:

MedTech: Devices, diagnostics, and therapeutics grounded in clinical science and regulatory pathways.

Competitive Landscape

Solution	Description	Accuracy	Population fit	Verdict
Manual Specialist	MOH forensic manually scores OPG using the atlas method. Cases referred from PDRM and JIM	Mean Absolute Error (1.5-3 years)	Partial My data	Days-subjective
F-DentEstLab (UM/Mohammad et al.)	Academic deep CNN for Malaysian paediatric OPG dataset. Semi automated, research-grade. Not commercialised or productised	Strong on children only	Malaysian Dataset	No product yet
KU Leuven/ EU AI tools	Academic AI models(DenseNet,Xception) validated on European/ Turkish populations.	Mean Absolute Error (1-2 years)	Non-My population	No MY fit
Our solution (UM IP)	ResNeXt-50 CNN fine-tuned on Malaysian multi-ethnic OPG dataset (Malay, Chinese,Indian). PDRM+ JIM deployment.	Mean Absolute Error (<2 yrs target)	MY-specific	Only product

Pricing Anchor

Dental Specialist Professional Fees in Malaysia: **RM 400 ~ RM 1,000 per hour**

Single dental age estimation case (includes Orthopantomogram (or OPG) review, report and court-ready documentation) takes **2 ~ 4 hours**:

RM 400 ~ RM 1,000 x 2 ~ 4 hours = **RM 800 ~ RM 4,000 (per case)**

Assuming 100 cases per month, that will be **RM 80,000 ~ RM 400,000**.

With our Professional Tier Subscription, **RM 2,500**,

RM 2,500 / 100 = RM 25 (per case)

Phase 1

Grant + Pilot (Year 1 & Year 2)

Funding Source

1. Government grant funding

Amount grant: RM 130,000
Purpose: Validate our product (Boost to TRL 5)
Timing: Apply month 1, receive month 4-5

2. Paid pilot engagements

Number of pilots: 2 (PDRM & JIM)
Amount grant for pilot: RM40,000 (PDRM) & RM30,000 (JIM)
Total amount grant from pilot: RM 70,000
Timing: Start from second year

Phase 1 — Grant + Pilot (Year 1 & Year 2)		
Period	Initiative	Detail
Year 1		
Month 1–2	Grant application + IP licensing agreement	Submit commercialisation grant. Sign IP licensing heads of agreement with UMCIE. Begin outreach to PDRM Forensic Science Division and JIM to secure letters of intent for pilots.
Month 3–4	Grant received	RM 130,000 grant disbursed. (1) expanding training dataset with Malaysian-specific OPG images (2) fine-tuning ResNeXt-50 on local population data (3) building preprocessing pipeline for DICOM format compatibility. Deliverable: Internal dataset registry documenting OPG image count, demographic breakdown, and data source agreements — required for MDA technical file later.
Month 5–8	Model validation (Aim to reach TRL 5)	Internal validation on held-out Malaysian OPG dataset. Targeted to mean absolute error below 2 years & produce a validation report, which can be an evidence document for pilots. Failure condition: If MAE target is not met, extend fine-tuning cycle by 4 weeks before proceeding to pilot contracts. Do not sign pilots without validation evidence.

Month 9–12	Pilot contracts + mobilisation fees	Finalise pilot agreements with 2 sites: PDRM Forensic Lab (RM 40K total) and JIM (RM 30K total). Each contract includes a RM 10K mobilisation fee collected on signing. Build deployment platform and non-technical UI in parallel. RM 20K mobilisation revenue bridges the pre-pilot cash gap. UI acceptance criteria (added): A forensic officer with no AI background can upload an OPG and receive a report without assistance within 3 minutes. Mandatory compliance step: Appoint a Data Protection Officer prior to Phase 1 deployment. OPG datasets are categorized as biometric sensitive information under the PDPA Amendment Act 2024. Our on-premises architecture directly addresses PDPA residency and MOH sovereignty mandates.
Year 2		
Month 13–19	Pilot 1 go live	PDRM - Forensic Lab pilots deployed. Service includes: system setup, staff training, 6-month supported usage period, and a performance report. The Forensic Lab provides real case data. This is your TRL 6 validation — real-world deployment with institutional users. Success criteria (added): MAE < 2.5 years on real forensic cases, system uptime > 95%, and a signed performance report from PDRM forensic division head. Subscription conversion (added): Month 19: PDRM pilot converts to RM 2,500/month Professional subscription upon signing performance report.
Month 17	MREC ethics application filed	File ethics approval application (MREC or institutional IRB) for the Month 20 clinical validation study. Filing at Month 17 — while the PDRM pilot is still running — ensures approval is in hand before Month 20 study start. Important: Do not wait for pilot results to begin ethics paperwork.
Month 20–22	Pilot 2 go live	JIM - Jabatan Imigresen Malaysia (legal authorities) pilot deployed. JIM provides data from immigration detention centres for age verification on undocumented individuals. Clinical validation study (added): Engage UMMC or Hospital Kuala Lumpur under a formal research agreement. Minimum 300 OPG cases with known ground-truth ages (Malay, Chinese, Indian, age range 10–70). Model output compared against clinician manual assessment as gold standard. Success criteria for TRL 7 (added): MAE < 2.5 years on held-out test set; sensitivity and specificity reported by age band; results reproducible by independent operator. Deliverable (added): Clinical study report (CSR) in MDA-compliant format — Annex 1 of the Class B submission.

Month 22–24	Regulatory dossier — MDA Class B pre-submission	Engage a local regulatory affairs consultant (MDA-registered) to prepare the technical file. Required components: (1) CSR from Month 20–22 study, (2) software lifecycle documentation per IEC 62304, (3) risk management file per ISO 14971, (4) intended use and labelling, (5) cybersecurity assessment. Submit pre-submission enquiry (PSE) to MDA — a free consultation step that confirms device classification and identifies dossier gaps before formal submission. This prevents a costly rejection at Month 28.
Month 23–24	Phase 1 closeout (Ready for phase 2)	Publish pilot results report. 3 case studies documented. Approach MOH procurement for national rollout discussion. Case study content: Each case study to include: case count processed, average turnaround time vs. manual baseline, cost per case comparison, and officer satisfaction rating. These form the evidence pack for MOH procurement conversations.
Summary TRL at Month 24: 7 (clinically validated) MDA pre-submission: Complete Clinical cases validated: 300+ OPGs Pilot sites confirmed: 3		
Risk: If MREC ethics approval takes longer than 6 weeks, Month 20 study start slips. Apply for ethics approval at Month 17 (before TRL 6 PDRM pilot ends) so approval and pilot completion overlap. Do not wait for pilot results to begin ethics paperwork.		

Phase 2

Converting pilots to subscribers.

THREE SUBSCRIPTION TIERS — WHAT WE SELL

1. Starter — small clinics and single-dept units

- Price:** RM999.90/month
- Who buys this:** Private dental clinics (1–3 branches), small district hospital pathology units, individual forensic practitioners.
- What they get:** Cloud-hosted access, up to 100 OPG reads/month, PDF report generation, email support, standard API access.
- Why this tier works:** Low commitment entry point for private sector. Easy to upgrade when volume grows. Pay-per-use overage at RM 10/read above 100.

2. Professional — government depts and hospital networks (Most Revenue)

- Price:** RM 2,499.90/month
- Who buys this:** PDRM Forensic Division, JIM regional offices, MOH district hospitals, dental chains (4+ branches). This is your Phase 1 pilot conversion tier.
- What they get:** Unlimited OPG reads, on-premises or cloud deployment option, case management integration (DICOM/HL7), priority support SLA (4hr response), quarterly performance reports, multi-user access (up to 20 users).

- d. **Why this tier works:** Matches government procurement budget cycles. Unlimited reads removes usage anxiety for high-volume departments. On-premises option satisfies MOH data sovereignty requirements.

3. Enterprise — national rollouts and ASEAN expansion

- a. **Price:** RM 119,999.90/year
- b. **Who buys this:** MOH master contract (all government hospitals), PDRM national deployment, Singapore/Indonesia forensic agencies, large hospital group (KPJ, Sunway Medical).
- c. **What they get:** Multi-site deployment under one contract, custom model fine-tuning for their population dataset, dedicated account manager, SLA with uptime guarantee, white-label option, quarterly in-person review, API rate limit removed.
- d. **Why this tier works:** One enterprise contract (e.g. MOH national) equals the revenue of 3 Professional sites. Target from Year 4 once MDA registration is complete and 2+ case studies are published.

Year 2 Bridge — TRL 6 to TRL 7 (Months 20–24)

Period	Initiative	Detail
TRL 6 → 7 Clinical validation study required for MDA Class B submission. This section is the explicit bridge. Without it, Month 28–33 MDA application has no evidentiary foundation.		
Month 20–22	Hospital-based clinical validation study — TRL 7 evidence generation	Engage a UM-affiliated hospital (UMMC or Hospital Kuala Lumpur) under a formal research agreement to conduct a prospective clinical validation study. This is distinct from the PDRM forensic pilot (TRL 6) — it must be in a licensed clinical setting with ethics board approval (MREC or institutional IRB). Study design: minimum 300 OPG cases with known ground-truth ages across all demographic groups (Malay, Chinese, Indian, age range 10–70). Model output compared against clinician manual assessment as gold standard. Success criteria for TRL 7: mean absolute error (MAE) < 2.5 years on the held-out test set; sensitivity and specificity reported by age band; results reproducible by an independent operator. Deliverable: clinical study report (CSR) in MDA-compliant format — this document is Annex 1 of the Class B submission.

Month 22–24	Regulatory dossier preparation — MDA Class B pre-submission	Engage a local regulatory affairs consultant (MDA-registered) to prepare the technical file. Required components: (1) CSR from Month 20–22 study, (2) software lifecycle documentation per IEC 62304, (3) risk management file per ISO 14971, (4) intended use and labelling, (5) cybersecurity assessment. Submit pre-submission enquiry (PSE) to MDA — a free consultation step that confirms the device classification and identifies any gaps in the dossier before formal submission. This prevents a costly rejection at Month 28. TRL 7 is confirmed when the MDA pre-submission feedback does not require additional clinical data. If it does, Month 25 buffer allows one corrective study cycle before formal submission. Parallel: file patent claims in Malaysia (if not yet granted) and begin PCT application to lock ASEAN IP position before the MDA submission makes the clinical data public.
TRL 7 → 7+ TRL 7 confirmed: system validated in intended operational environment (licensed hospital), clinical evidence documented, regulatory dossier complete. MDA Class B formal submission can now proceed at Month 28.		
Summary TRL at Month 24: 7 (clinically validated) MDA pre-submission: Complete Clinical cases validated: 300+ OPGs Regulatory dossier: Draft complete		
Risk: If MREC ethics approval takes longer than 6 weeks, Month 20 study start slips. Apply for ethics approval at Month 17 (before TRL 6 PDRM pilot ends) so approval and pilot completion overlap. Do not wait for pilot results to begin ethics paperwork.		
Year 3 — Domestic Dominance (Months 25–36)		
Period	Initiative	Detail
Month 25–27	Subscription launch — convert pilots to paid subscribers	Formally launch subscription product. Offer Phase 1 pilot sites (PDRM, JIM) a discounted Year 1 rate (RM 960/month, 20% loyalty discount) to incentivise conversion. Target: all 3 pilot sites convert. Publish PDRM and JIM pilot performance reports as public case studies. Begin ePerolehan (government procurement portal) vendor registration.

Month 28–33	Government channel — 10 new sites via MOH + formal MDA submission	Submit formal MDA Class B registration application using the clinical study report (CSR) from Month 20–22 and the regulatory dossier finalised at Month 24. This is the earliest credible submission date — all required evidence is now in hand. Use JIM reference and TRL 7 clinical evidence to approach MOH Forensic Odontology unit and district hospital pathology network. Target: 10 new Professional tier subscriptions by Month 33. Running ARR at Month 33: 13 sites x RM 2,499.90 = RM 32,499/month (RM 389,988 ARR). MDA Class B review takes 12–18 months. Registration expected to complete Month 40–46 — aligns with Year 4 MOH national tender window.
Month 34–36	Private dental channel opens — 7 new Starter tier sites	Activate dental distributor partnership (Henry Schein Malaysia, Dentsply Sirona dealer network) as a channel. Dental chains close faster — no government procurement cycle. Target: 7 Starter tier dental sites at RM 999.90/month. Year 3 close: 20 sites total. ARR: 13 Professional (RM 389,988) + 7 Starter (RM 83,992) = RM 473,980.

Summary | Year 3 Sites: 20 total | ARR: RM 473,980 | Gross Margin: ~72% | MDA Status: Under review (Month 28 submission)

Risk: MDA registration takes 12–18 months. The Month 28 submission is only possible because TRL 7 clinical evidence was generated in Months 20–22. If clinical study was skipped or results were insufficient, this submission date is invalid and Year 4 hospital procurement is blocked.

Year 4 — Domestic Scale + First ASEAN Foothold (Months 37–48)

Period	Initiative	Detail
Month 37–42	MOH national tender + MDA registration completes	MDA Class B registration completes (Month 40–46 window) — unlocks full public hospital procurement without research exemption. The clinical evidence from Month 20–22 is the basis of registration approval. Submit MOH national tender for forensic pathology digitisation across all district hospitals. Target: MOH master contract covering 15 district hospitals (Sabah, Sarawak, Kelantan prioritised) signed by Month 42. Value: 15 x Professional = RM 449,982/yr.

		PDRM national rollout signed: 12 regional forensic divisions at Professional tier = RM 359,986/yr.
Month 43–45	Singapore HSA pilot submission	Submit HSA (Health Sciences Authority) regulatory clearance application for Singapore. The MDA-registered status in Malaysia plus the published clinical study report significantly accelerates HSA review. Engage 1–2 Singapore dental school partners for validation dataset top-up (local population). Target: 2 Singapore Professional tier pilot sites by Month 45 (research status while HSA review is in progress). Value: RM 59,998/yr. Publish 2nd peer-reviewed validation paper in indexed forensic journal — activates credibility with ASEAN forensic bodies.
Month 46–48	KPJ or Sunway Medical Enterprise contract + Indonesia pilot	First private hospital group enterprise contract (KPJ Healthcare or Sunway Medical). Enterprise tier: RM 119,999.90/yr covers all group hospitals under one contract. Activate Indonesia Puslabfor (National Police Forensics Lab) pilot — 3 cities (Jakarta, Surabaya, Medan), Professional tier. Year 4 close: 55+ sites across MY, SG, ID.
Summary Year 4 Sites: 55+ (MY+SG+ID) ARR: RM 1.8M Gross Margin: ~76% MDA Status: Registered — full hospital procurement unlocked		
Risk: ASEAN regulatory timelines are unpredictable. If HSA Singapore delays beyond Month 45, redirect to Thai or Philippine forensic labs which accept CE-equivalent pathway. The MDA registration is reusable as reference evidence for most ASEAN regulators.		
Year 5 — Platform & Licensing (Months 49–60)		
Period	Initiative	Detail
Month 49–52	White-label licensing deal with PACS/RIS vendor	Close a white-label OEM agreement with a regional PACS or RIS vendor (target: Agfa HealthCare, Infinitt, or Telerad). The OPG age estimation module ships embedded inside their imaging platform. This opens hospitals that already have vendor relationships — no direct sales cycle needed. Licensing fee: RM 150K–300K/yr per vendor agreement. Vendor handles regulatory clearance in their markets; you provide the model API and update cadence.

Month 52–55	Thailand and Philippines rollout	Use Indonesia and Singapore references to approach Thailand Department of Special Investigation (DSI) and Philippine NBI forensic units. Target: 8 Professional tier sites across TH + PH by Month 55. Value: RM 239,990/yr. Population model fine-tuning for Thai and Filipino dental datasets — strengthens competitive moat.
Month 55–57	Population model marketplace launch	Launch API marketplace allowing third-party researchers and diagnostic labs to fine-tune and host their own regional model variants on your infrastructure. Target: 5 paying research institutions in Year 5. Estimated revenue: RM 600K. UM royalty structure renegotiated to reflect commercial scale. Engage UMCIE for restructured IP licensing terms.
Month 58–60	Series A / strategic acquisition positioning	ARR and ASEAN footprint (6 countries, 120+ sites, 2 white-label partners) used as leverage for Series A fundraise (target: RM 8–12M) or strategic acquisition approach. Ensure: model retrained annually with new clinical data (data moat), all contracts have minimum 12-month lock-in, UM IP licence fully documented for due diligence. Appoint dedicated account manager + regional BD hire (Singapore-based) to sustain ASEAN pipeline.
Summary Year 5 Sites: 120+ direct + white-label ARR: RM 8.5M Gross Margin: ~81% Revenue Streams: SaaS + Licensing + Marketplace		
Risk: Moat check: if a PACS vendor builds this natively, defensibility rests on UM-exclusive training dataset + patented method + published clinical validation studies. Keep dataset proprietary, update the model annually, and maintain researcher relationships at UM.		

Go-To-Market Plan

Our go-to-market strategy follows a three-channel, phased approach — starting with direct government engagement where the pain is highest and the budget is real, then expanding through institutional procurement channels, and finally activating private sector distribution once regulatory clearance is in place.

Channel 1 — Direct Government Sales (Year 1–2)

The first and primary channel is direct outreach to PDRM Forensic Science Division and Jabatan Imigresen Malaysia (JIM). Both agencies have documented, high-volume caseloads that require age verification and no adequate existing solution. Entry is through Letters of Intent secured during Month 1–2 of the roadmap, converting to paid pilot agreements at Month 9–12. The government procurement pathway is real and budget-allocated — PDRM and JIM operate under the Ministry of Home Affairs (MOI), which has established forensic digitisation as a priority. This channel does not require MDA registration to initiate — pilots can proceed under research and evaluation status, which is standard for government technology assessments in Malaysia.

Channel 2 — ePerolehan Government Portal (Year 3)

Once pilot results are published as case studies and the MDA Class B pre-submission is complete, we register as a vendor on ePerolehan — Malaysia's official government procurement portal. This opens the broader MOH district hospital network, forensic odontology units, and additional PDRM regional divisions without needing a direct sales relationship at each site. Target: 10 new Professional tier subscriptions by Month 33 sourced through this channel. ePerolehan registration begins at Month 25 in parallel with subscription launch.

Channel 3 — Dental Distributor Partnership (Year 3–4)

The private dental sector (dental chains, private hospitals) operates on shorter procurement cycles than government — no tender process, no ethics board. To reach this segment efficiently, we activate a channel partnership with established dental equipment distributors in Malaysia, targeting networks such as Henry Schein Malaysia and Dentsply Sirona dealer channels. Distributors already have trusted relationships with clinic owners and dental chains, and can bundle our software subscription alongside hardware purchases. This channel targets the Starter tier (RM 999.90/month) and opens 7+ sites by Month 34–36 without a direct sales team.

Channel 4 — White-Label OEM Licensing (Year 5)

At scale, the highest-leverage GTM move is embedding our OPG age estimation module inside existing PACS (Picture Archiving and Communication System) and RIS (Radiology Information System) platforms used by hospitals across ASEAN. Target vendors include Agfa HealthCare, Infinitt, and Telerad. Under a white-label OEM agreement, the vendor handles sales, deployment, and regulatory clearance in their markets — we provide the model API and annual update cadence for a licensing fee of RM 150K–300K per vendor agreement per year. This channel removes the direct sales bottleneck entirely for ASEAN expansion.

GTM Summary

Phase	Channel	Target Customer	Tier	Timeline
Phase 1	Direct government sales	PDRM, JIM	Professional	Year 1–2
Phase 2	ePerolehan portal	MOH hospitals, PDRM regional	Professional	Year 3
Phase 2	Dental distributor network	Private dental chains	Starter	Year 3–4
Phase 3	White-label OEM licensing	ASEAN hospital networks	Enterprise	Year 5

Scaling Potential

Our scaling follows three sequential layers, each unlocked by the evidence built in the layer before it.

Layer 1 — Domestic Scale (Year 3–4)

MDA Class B registration at Month 40–46 unlocks the full MOH district hospital network, all PDRM regional divisions, and JIM detention centres nationwide through the ePerolehan government procurement portal — no direct sales needed at each site. Target: 55 active sites by end of Year 4, generating RM 1.8M ARR at 76% gross margin.

Layer 2 — ASEAN Expansion (Year 4–5)

Malaysia's MDA registration and published clinical studies serve as reference evidence for neighbouring regulators, shortening approval timelines in each new market. Expansion sequence:

- **Singapore** — Month 43–45, HSA pilot submission
- **Indonesia** — Month 46–48, National Police Forensics Lab (Puslabfor)
- **Thailand & Philippines** — Month 52–55, DSI and NBI forensic units

Each market entry includes population-specific model fine-tuning, deepening accuracy and widening the competitive moat simultaneously.

Layer 3 — Platform & Licensing (Year 5)

White-label OEM agreements with regional PACS/RIS vendors (Agfa HealthCare, Infinitt, Telerad) embed our module inside imaging platforms already deployed across ASEAN hospitals — opening hundreds of sites through

one vendor relationship at RM 150K–300K per agreement per year. An API marketplace further allows research institutions to fine-tune regional model variants on our infrastructure.

By Year 5: 6 countries, 120+ sites, 2 white-label partners, RM 8.5M ARR across SaaS, licensing, and marketplace revenue streams.

What keeps this defensible: The UM-exclusive Malaysian training dataset, published peer-reviewed validation studies, annual model retraining with new clinical data, and patent PI 2025003749 — none of which a new entrant can replicate quickly.

Unit Economics

1) Customer Acquisition Cost (CAC)

Estimated Cost: **RM 18,000**

- Demo costs, travel and procurement process time
- Cost grant by government

2) Lifetime Value (LTV)

- Monthly revenue = RM 2,499.90
- Annual revenue = RM 2,499.90 x 12 = RM 29,998.80

Contracts renew in 3 Years

- RM 29,998.80 x 3 = **RM 89,996.40**

3) LTV:CAC ratio

- CAC = RM 18,000
- LTV = RM 89,996.40
- LTV:CAC = RM 89,996.40 / RM 18,000 = **4.99**
- Government contracts are high value and low churn rate

4) Payback

- Monthly revenue = RM 2499.90
- Monthly direct cost = RM 2,499.90 x 20% = RM 499.98
- Monthly gross profit = RM 2,499.90 - RM 499.98 = RM 1999.92
- Payback Period = RM 18,000 / RM 1,999.92 = **9 Months**

5) Gross Margin %

- Monthly gross profit = RM 1,999.92
- Monthly revenue = RM 2,499.90
- Gross margin = (RM 1,999.92 / RM 2,499.90) x 100 = **80%**

Cost Structure

COST STRUCTURE

Year 1–2 operating costs

Government commercialisation grant	+RM 130,000
Dataset expansion & OPG licensing	RM 30,000–50,000
ResNeXt-50 fine-tuning compute	RM 15,000–25,000
DICOM preprocessing pipeline build	RM 10,000–15,000
Cloud hosting & infrastructure	RM 12,000/yr
UI / deployment platform build	RM 15,000–20,000
IP licensing fee to UMCIE	ongoing royalty
Regulatory affairs consultant (MDA)	RM 20,000–40,000
MREC ethics application & CSR study	RM 15,000–30,000
Pilot mobilisation revenue (bridge)	+RM 20,000
RM 10K × 2 sites on contract signing	

Ongoing direct cost breakdown

Monthly direct cost per customer	RM 500
Monthly revenue per customer	RM 2,500
Monthly gross profit per customer	RM 2,000
Annual direct cost per customer	RM 6,000
Annual gross profit per customer	RM 24,000
Gross margin %	80%

Gross margin trajectory

Yr 1–2		~70%
Yr 3		~72%
Yr 4		~76%
Yr 5		~81%

SUBSCRIPTION TIERS — WHAT WE SELL

Starter

Private clinics

RM 999.90/mo

Up to 100 OPG reads/mo · RM 10 overage

Private dental clinics (1–3 branches), small district units, individual practitioners

Professional

Most revenue

Pilot conversion tier

RM 2,499.90/mo

Unlimited reads · on-premises or cloud · 4hr SLA · up to 20 users

PDRM Forensic, JIM regional, MOH district hospitals, dental chains (4+ branches)

Enterprise

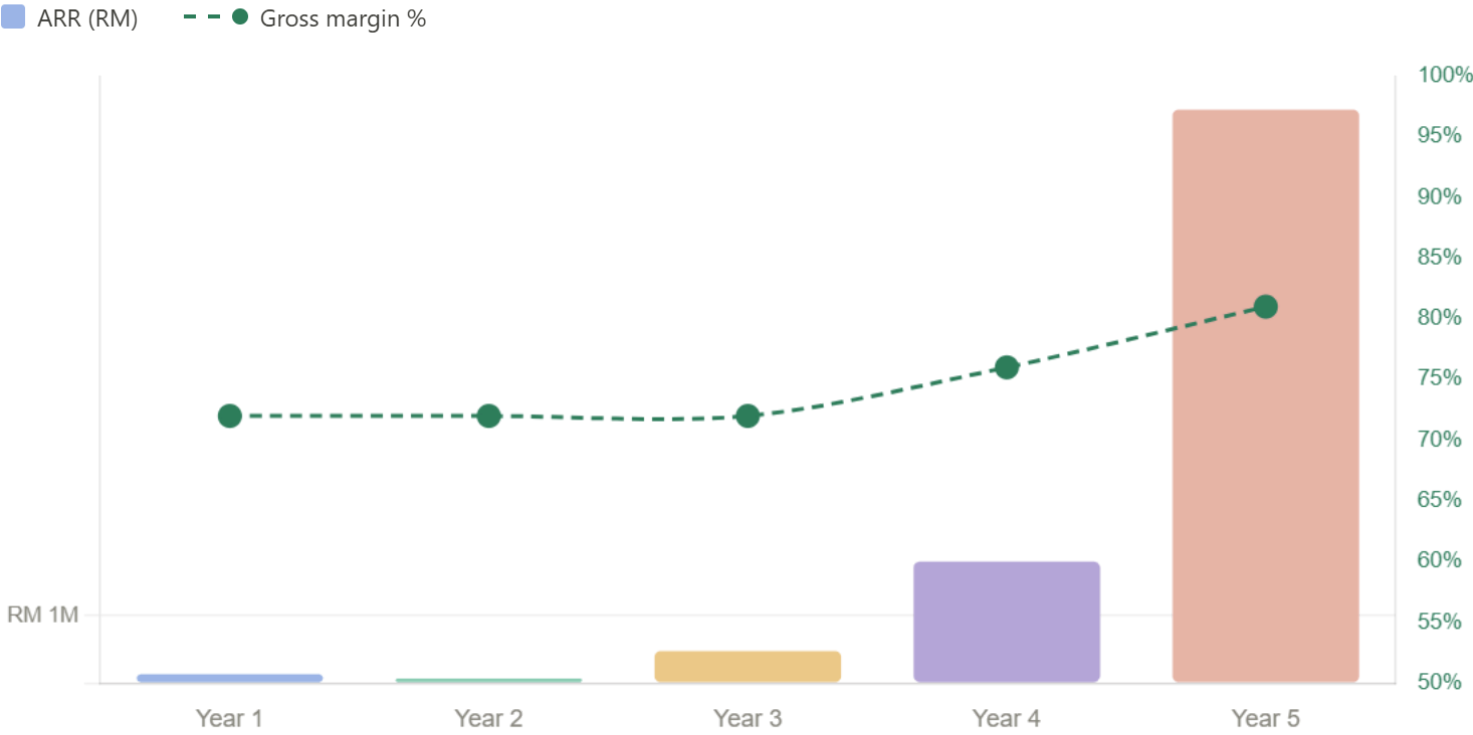
Yr 4+ target

RM 119,999.90/yr

Multi-site · custom model fine-tuning · white-label · dedicated account manager

MOH national contract, PDRM national rollout, KPJ/Sunway Medical, ASEAN agencies

ARR GROWTH TRAJECTORY



REVENUE STREAM COMPOSITION BY PHASE

Phase 1 (Yr 1–2)

Grant · Pilot fees · Mobilisation · Early subscriptions

Phase 2 (Yr 3)

Professional SaaS (primary) · Starter SaaS (secondary)

Phase 3 (Yr 4)

Professional · Enterprise · ASEAN pilot fees

Phase 4 (Yr 5)

SaaS · White-label licensing · API marketplace

KEY COST & REVENUE RISKS

MREC ethics approval delay

File Month 17 (before PDRM pilot ends) to prevent Month 20 study slip. Never wait for pilot results.

MDA registration timeline

12–18 month review. Month 28 submission unlocks MOH national tender only if TRL 7 clinical evidence is in hand.

MAE validation failure

If MAE >2 yrs at Month 5–8, extend fine-tuning 4 weeks. Do not sign pilots without validation evidence.

ASEAN regulatory unpredictability

If HSA Singapore delays beyond Month 45, redirect to Thai or Philippine forensic labs (CE-equivalent pathway).

Team Structure

Our team of four brings together complementary technical foundations from Software Engineering and Computer Science, collectively covering the full stack required to build, validate, and commercialise an AI-powered MedTech product — from model development and system architecture to business strategy and project execution.

Chang Yi Hern — AI & Technology Lead *BSc Computer Science, Sunway University*

Yi Hern leads the core AI development workstream, overseeing the fine-tuning of the ResNeXt-50 convolutional neural network on Malaysian OPG datasets, model validation, and inference pipeline architecture. His Computer Science foundation provides the theoretical grounding in machine learning, image processing, and algorithm optimization that the technical core of this IP demands. Within the project, Yi Hern is responsible for model accuracy targets (MAE < 2 years), the preprocessing pipeline, and ensuring the system meets the UI acceptance criteria — a forensic officer with no AI background completes a full report within 3 minutes.

Wong Ming Ze — Systems & Platform Lead *BSc Software Engineering, Sunway University*

Ming Ze leads the development of the deployment platform — the software layer that sits between the AI model and the end user. His Software Engineering background covers system design, database architecture, cloud infrastructure, and full-stack development, all of which are directly required to build a production-ready platform that integrates with hospital imaging systems via DICOM and HL7 protocols. Within the project, Ming Ze is responsible for platform stability, cybersecurity compliance, and achieving the uptime target of above 95% during pilot deployments.

Bryan Tan Chun Xing — Business & Commercialisation Lead *BSc Software Engineering, Sunway University*

Bryan leads the commercialisation strategy, revenue model, and stakeholder engagement workstream. His Software Engineering background gives him the technical literacy to translate complex AI capabilities into clear value propositions for non-technical buyers — a critical skill when selling to government procurement officers at PDRM and JIM who evaluate outcomes, not algorithms. Within the project, Bryan is responsible for the grant application process, pilot contract negotiations, pricing strategy, and the ePerolehan vendor registration pathway.

Chin Yoong Khang — Data & Validation Lead *BSc Software Engineering, Sunway University*

Yoong Khang leads the data pipeline and validation workstream — the foundation on which the entire accuracy claim rests. His Software Engineering background covers data engineering, statistical analysis, and system testing, directly applicable to the tasks of OPG dataset collection, image standardisation, labelling, augmentation, and performance monitoring. Within the project, Yoong Khang is responsible for maintaining the internal dataset registry, producing the validation report at Month 8, and ensuring the clinical study report (CSR) meets MDA-compliant format requirements for the Class B submission.

Why this team for this IP

All four members are students at Universiti Malaya — the same institution that developed and owns the patent (PI 2025003749). This gives the team direct access to the IP inventors, the UM research dataset, and the UMCIE commercialisation office — advantages that an external team cannot replicate. The combination of Computer Science and Software Engineering across all four members means every technical component of the roadmap — model development, platform build, data pipeline, and system integration — is covered internally without outsourcing core IP work to third parties.

AI & Development Team

Responsible for designing, developing, and training the AI-powered age estimation system using deep learning and convolutional neural networks (CNN). The team develops the software platform, integrates the AI prediction engine, manages databases and cloud systems, and ensures smooth automation from image upload to age prediction output. They also improve system accuracy, optimize performance, fix technical issues, and maintain cybersecurity and system reliability.

Data Analytics Team

Responsible for collecting, organizing, preprocessing, and analyzing dental orthopantomogram (OPG) image datasets used for AI training and validation. The team performs data cleaning, image standardization, labeling, augmentation, and statistical analysis to ensure high-quality datasets. They also monitor model performance metrics such as prediction accuracy and mean absolute error (MAE), helping improve the reliability and generalization capability of the AI system.

Forensic & Medical Validation Team

Consists of forensic specialists, dentists, orthodontists, radiologists, and medical advisors who validate the clinical and forensic reliability of the system. The team compares AI-generated age predictions with traditional manual assessment methods to ensure accuracy and consistency. They conduct internal validation testing using Malaysian OPG datasets, prepare validation and evidence reports for pilot studies, and ensure the technology complies with medical, ethical, and forensic standards required for future regulatory approval and commercialization.

Business & Finance Team

Responsible for managing the project's financial planning, budgeting, and commercialization strategy. The team estimates development costs, manages grants and funding opportunities, conducts market analysis, evaluates competitors, and identifies business opportunities in the MedTech and forensic sectors. They also handle partnerships with hospitals, forensic laboratories, universities, and government agencies, while developing pricing strategies, subscription models, licensing opportunities, and long-term revenue plans for the IP.

Project Management Team

Responsible for overseeing the overall project execution, coordination, and strategic planning. The team manages project timelines, milestones, task delegation, communication between technical and medical teams, and documentation progress. They monitor risks, ensure deliverables are completed according to schedule, coordinate meetings and reporting activities, and ensure that all teams work collaboratively toward successful prototype development, pilot preparation, and future commercialization goals.

The Ask

We are seeking **RM 200,000** in total funding across two streams to bring our AI-powered dental OPG age estimation system from its current TRL 3 to a clinically validated, pilot-ready product by Month 24.

Stream 1 — Commercialisation Grant: RM 130,000

To be applied for in Month 1 from government grant programmes (MTDC, UMCIE commercialisation fund, or equivalent). These funds are allocated specifically to:

- Expanding and annotating the Malaysian multi-ethnic OPG training dataset (Malay, Chinese, Indian demographics)
- Fine-tuning the ResNeXt-50 model on local population data to achieve MAE below 2 years
- Building the preprocessing pipeline for DICOM format compatibility with hospital imaging systems
- Developing the non-technical deployment UI (target: forensic officer with no AI background completes a report within 3 minutes)
- Appointing a Data Protection Officer to ensure PDPA Amendment Act 2024 compliance before any live deployment

Deliverable at Month 8: Internal validation report confirming MAE < 2 years on held-out Malaysian OPG dataset — the evidence document required before any pilot agreement is signed.

Stream 2 — Paid Pilot Contracts: RM 70,000

Two institutional pilot agreements secured in Year 2:

- PDRM Forensic Science Division — RM 40,000 (includes RM 10,000 mobilisation fee on contract signing)
- Jabatan Imigresen Malaysia (JIM) — RM 30,000 (includes RM 10,000 mobilisation fee on contract signing)

These are not grants — they are paid service contracts where each institution receives system deployment, staff training, 6 months of supported usage, and a performance report. The RM 20,000 combined mobilisation fees collected at Month 9–12 bridge the pre-pilot cash gap before grant funds are fully deployed.

What this funding unlocks:

Milestone	Timeline	Outcome
Model validated (MAE < 2 yrs)	Month 8	Pilot contracts signable
PDRM pilot live	Month 13	TRL 6 achieved
MREC ethics application filed	Month 17	Clinical study approved in time
JIM pilot live + clinical study	Month 20–22	TRL 7 achieved
MDA Class B pre-submission	Month 22–24	Regulatory pathway opened
Subscription launch	Month 25	First recurring revenue

What we are NOT asking for is funding to cover ongoing operating costs or salaries. Every ringgit requested is tied to a specific deliverable. If the MAE target at Month 8 is not met, we extend the fine-tuning cycle before proceeding — no pilot contracts are signed without validation evidence in hand.

By Month 24, this investment produces: 3 paying pilot sites, a clinically validated product (300+ OPG cases), a completed MDA pre-submission dossier, and a subscription product ready to launch — positioning the business for Year 3 domestic scale without additional external funding.